Saul Cooperman

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Summary

Software Engineer focused on building reliable, high-performance distributed systems in C++ and Python.
Currently developing latency-sensitive middleware at Bloomberg L.P., supporting thousands of real-time services.
Strong background in multithreading, distributed systems, and cross-team engineering.

Experience

Bloomberg L.P., Software Engineer — London, UK — Sept 2024 – present

- Core maintainer of a proprietary distributed messaging platform and gateway infrastructure processing 300B+ messages daily with 99.999% uptime, serving as mission-critical infrastructure across the firm.
- Lead development of high-performance, multithreaded C++ and Python SDKs with native bindings used by 17,000+ internal microservices across diverse production environments.
- Increased throughput of a high-performance multithreaded Python library with C++ bindings by 35% by building benchmarking and stress-testing infrastructure, identifying bottlenecks, and leading optimizations involving CPython internals and Python/C++ interoperability.
- Architect and own a schema-driven Python code generation framework adopted by 3,000+ production services (up from ~700), enabling versioned package distribution at enterprise scale.

Bloomberg L.P., Software Engineer Intern — London, UK — June 2023 – Sept 2023

- Built a system to analyse live encoded requests and evaluate backward-incompatible changes to the schema which encoded them.
- Enabled teams to evolve message schemas comfortably.

Education

BSc — **University of Leeds**, Computer Science — Leeds, UK — Sept 2021 – July 2024

- First Class Honours.
- Final Year Project (First): A neuromechanical model of C. elegans steering behaviour integrating sensory and neural mechanisms, optimized using evolutionary algorithms.

A-Level / Pre-U — **City of London School**, Mathematics, Further Mathematics, Physics, Chemistry — London, UK — Sept 2019 – July 2021

- Mathematics/Further Mathematics **D1** (**A****).
- Awarded “The Worshipful Company of Needlemaker’s Prize for IT and Computing”.

Personal Projects

PyStack subinterpreter support (open-source Python debugging tool)

- Contributed subinterpreter support, enabling accurate reconstruction of both Python and native (C) stack frames across multiple subinterpreters within a single process.

Generic Compilation Database Generator

- Reimplemented the C++ tool Bear in Rust, generating compilation database for Clang tooling by intercepting Linux build processes via LD_PRELOAD, syscall hooking, and a Unix Domain Socket server with Protobuf for structured data exchange.

TFL Times - Real-time Transit App

- Built a React frontend backed by a Python asyncio WebSocket server, including a fully custom typed TTL caching layer with extensive test coverage to minimise external API calls and latency.

Overengineered Tic-Tac-Toe

- Implemented a Tic-Tac-Toe engine in C++ with Python bindings via pybind11, using bitboards.
- Explored ARM NEON SIMD intrinsics for board evaluation and low-level perf optimizations.

Open source contributions to Pystack, Cython, CPython, BDE